R22

Code No.: 22ITC10

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E./B.Tech V Sem (Main) Examination December – 2024

(Common to CSE/IT/AI&DS)

Computer Networks

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A** at one place in the same order and **Part–B (Internal Choice)**

Part - A

(5Q X 2M = 10 Marks)

			M	CO	BT
1		How does a twisted pair cable reduce interference?	(2)	1	3
2		Why is error detection important in the Data Link Layer?	(2)	2	2
3		What is Network Address Translation (NAT)?	(2)	3	1
4		What are two essential elements of a transport protocol?	(2)	4	1
5		What is a domain resource record in the context of DNS?	(2)	5	1

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Define the following network topologies: bus, ring, star, and mesh. Provide one example of where each might be used.	(5)	1	1
	(b)	How would you analyze the four layers of the TCP/IP reference model and explain the role of each layer in facilitating communication across networks?	(5)	1	3
		(OR)			
7	(a)	Discuss the advantages of fiber optic cables? Explain with a neat sketch.	(5)	1	2
	(b)	Identify the functions of the physical and data link layers in the OSI model.	(5)	1	1
8	(a)	How would you apply the process of framing in the Data Link Layer to differentiate between various types of data, and what techniques could be used to implement this process effectively?	(5)	2	3
	(b)	Explain the concept of Carrier Sense Multiple Access (CSMA) and its variants.	(5)	2	3
		(OR)			
9	(a)	Explain how the cyclic redundancy check (CRC) technique detects errors in transmitted data. Provide an example of how CRC works in practice.	(5)	2	3
	(b)	Explain the operation of the Go-Back-N protocol in reliable data transmission.	(5)	2	1
10	(a)	Discuss the design issues in the Network Layer of the OSI model?	(5)	3	2
	(b)	Analyze how flooding works and why it can lead to inefficiencies in a network.	(5)	3	4
		(OR)			

11	(a)	Analyze how the Network Layer interacts with the Data Link Layer and Transport Layer in the OSI model. Discuss the importance of these interactions.	(5)	3	4
	(b)	How IPv6 addresses the problem of IP address exhaustion faced by IPv4? Explain.	(5)	3	3
12	(a)	Compare and contrast TCP and UDP Protocols.	(5)	4	2
	(b)	Explain the elements of transport layer	(5)	4	1
		(OR)			
13	(a)	Explain connection management in transport layer	(5)	4	1
	(b)	Draw the UDP header and explain all the fields	(5)	4	2
14	(a)	Examine the key components of the Domain Name System (DNS) and assess the specific role each component plays in ensuring the system's efficient operation.	(5)	5	4
	(b)	Explain about streaming audio	(5)	5	1
		(OR)			
15	(a)	What are the major components in E-mail system? Explain them.	(5)	5	1
	(b)	Write short notes on World Wide Web	(5)	5	1

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